EDS 222 HW1

Foundations of inference

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

1. Conduct a permutation test

Conduct a permutation test to investigate the following question:

> Do homes built pre-1970 have higher average lead soil concentrations?

To perform this test, you'll need the `exide-property-covariates.csv` file found in `/courses/EDS222/data`. This file contains the year each property was built.

> **Note**
>
> The question is about *average lead soil concentrations*. You'll need to first get the average lead soil concentration for each property, then join your result to the property covariate data to get the year the house was built.
> Null hypothesis: There is no difference in lead concentration in homes built pre-1970 and homes built post-1970.
> Alternate hypothesis: Houses built pre-1970 have higher levels of lead concentration than

those built post-1970.

```
# Reading in data
covariate <- read.csv("data/exide-property-covariates.csv")

soilcores <- read.csv("data/exide-soil-cores (1).csv")
```

```
# Summarizing average lead concentration by property
propconc <- soilcores |>
    summarise(meanconc = mean(pb_ppm, na.rm= TRUE), .by = property_id)
```

```
# Merging two dfs

everything <- merge(propconc,covariate, by = "property_id")
```

```
# Creating column for pre post 1970 build

everything <- everything |>
    mutate(before1970 = year_built <1970)

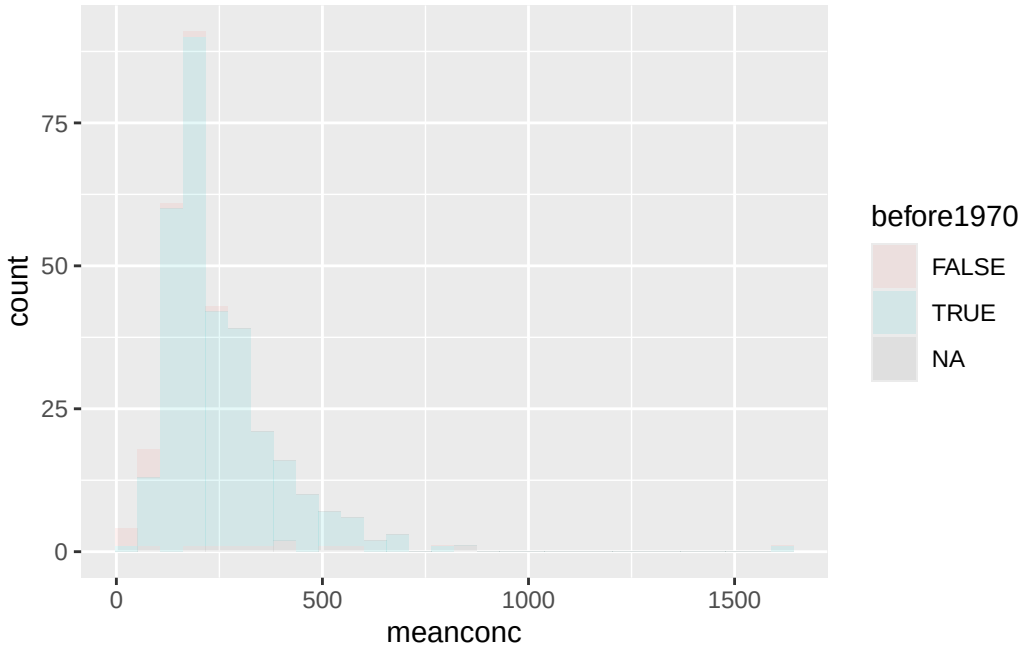
observedstat <- everything |>
    summarise(mean = mean(meanconc),.by = before1970)

oldnewdif <- observedstat[1,2] - observedstat[2,2]
```

```
# Visualizing lead concentration in houses by build date

everything |>
    ggplot(
        aes(x = meanconc, fill = before1970)
    )+
        geom_histogram(alpha = 0.1)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
# Sampling one time

samplstat <- everything |>
    mutate(sampleid = sample(before1970)) |>
    summarise(mean = mean(meanconc),.by = sampleid)

sampdif <- samplstat[1,2] - samplstat[2,2]
```

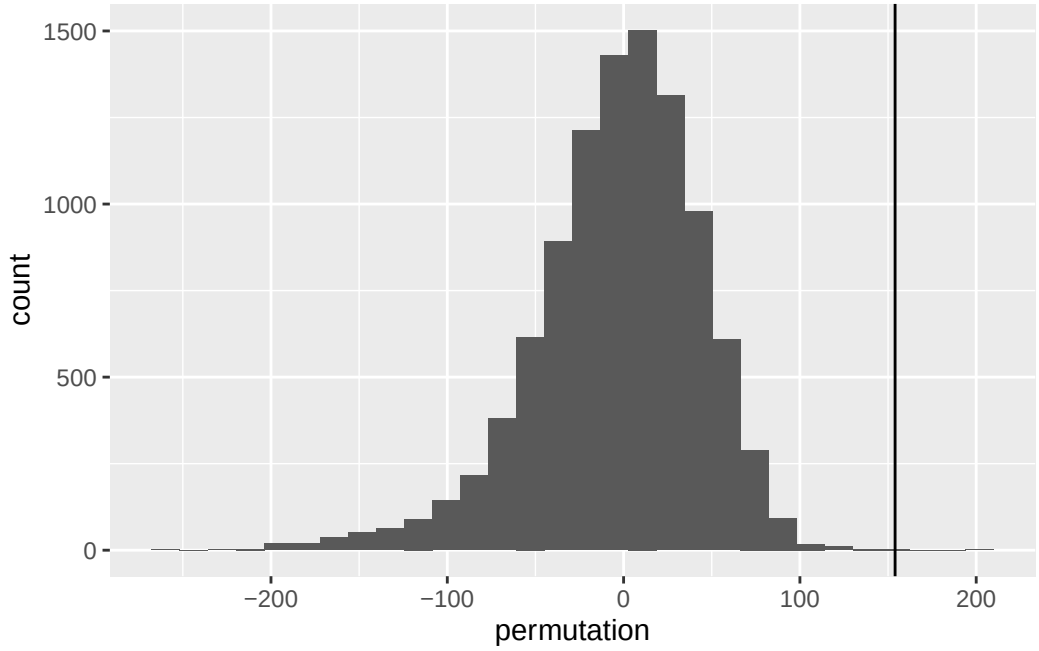
```
# Rerunning sampling 10000 times
permutation <- replicate(n = 10000,
{
   samplstat <- everything |>
    mutate(sampleid = sample(before1970)) |>
    summarise(mean = mean(meanconc),.by = sampleid)

sampdif <- samplstat[1,2] - samplstat[2,2]
}
)
permutation <- data.frame(permutation)
```

```r
# Visualize null distribution

permutation |>
    ggplot(
        aes(x = permutation)
    )+
        geom_histogram() +
        geom_vline(xintercept = oldnewdif)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
# Calculating p value

mean(permutation >= oldnewdif)
```

```
[1] 5e-04
```

The p value is 0. This means that 0 percent of the data from the permutation is greater than our observed difference between lead concentration in pre70s houses and post70s houses.

2. Estimate a confidence interval

Estimate a confidence interval to investigate the following question:

> How much is the difference in average lead soil concentrations between properties within 1km of a freeway and those farther away?

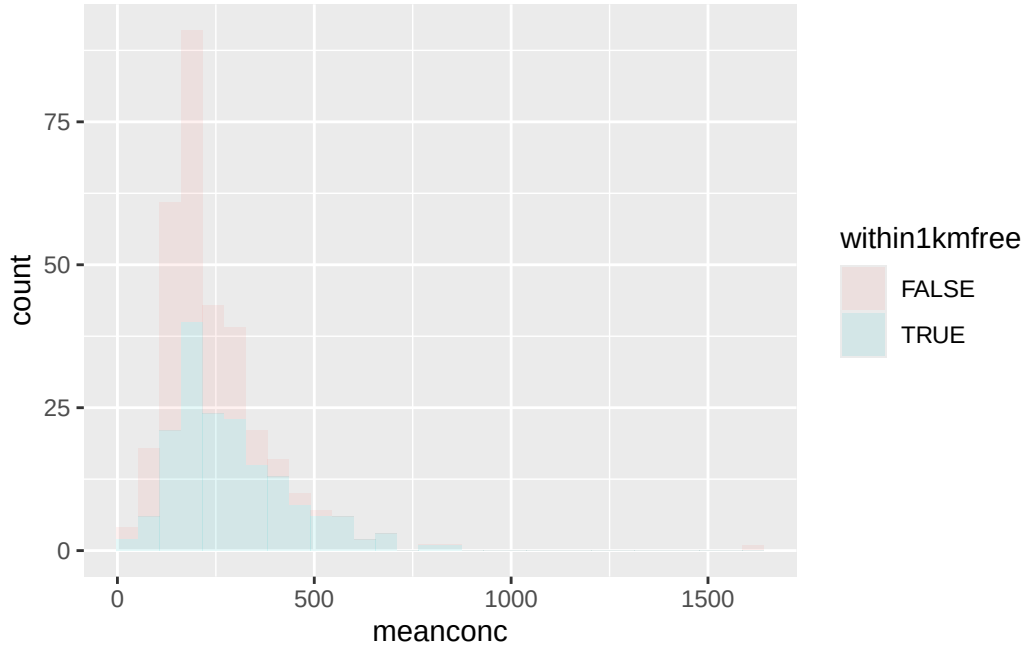
```
# Creating observed stat
everything <- everything |>
    mutate(within1kmfree = (dist_to_freeway_m < 1000))

freeway <- everything|>
    summarise(concbyprox = mean(meanconc), .by = within1kmfree)

sampstatfree <- freeway$concbyprox[1] - freeway$concbyprox[2]
```

```
# Visualizing lead concentration in houses by proximity to freeway
everything |>
    ggplot(
        aes(x = meanconc,fill = within1kmfree)
    )+geom_histogram(alpha = 0.1)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```r
# Bootstrapping, which involves selecting random rows to draw new samples

bootstrapgroup <- everything |>
    slice_sample(replace = TRUE, n = nrow(everything)) |>
    summarise(mean = mean(meanconc), .by = within1kmfree)

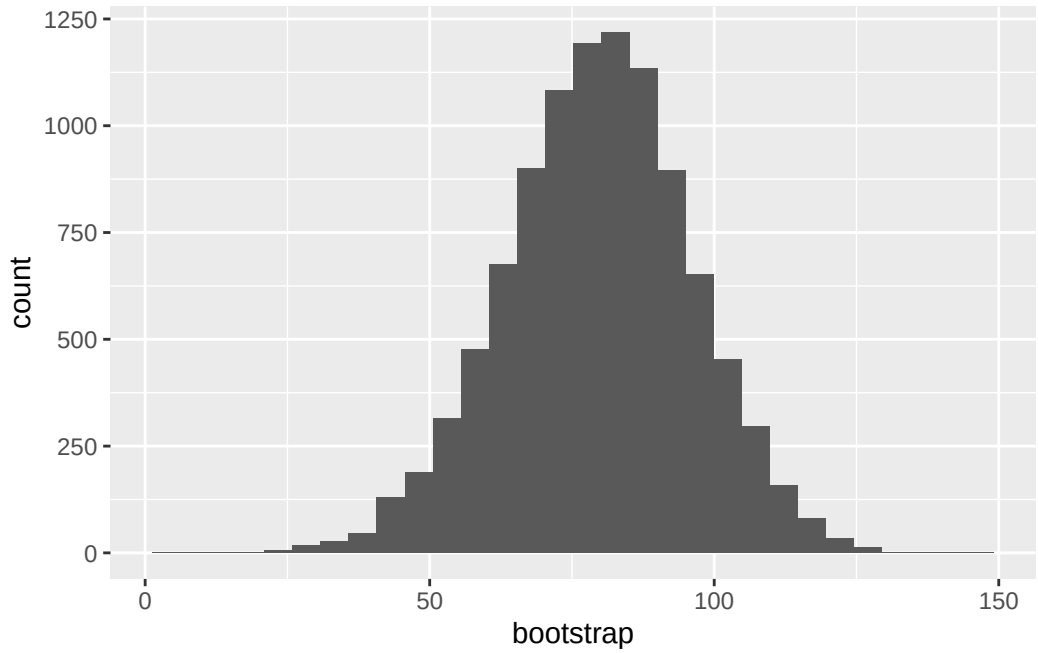
bootstrapstat <- bootstrapgroup$mean[1] - bootstrapgroup$mean[2]
```

```r
# Replicating 10000 times
bootstrap<- replicate(n = 10000,
{
    bootstrapgroup <- everything |>
    slice_sample(replace = TRUE, n = nrow(everything)) |>
    summarise(mean = mean(meanconc), .by = within1kmfree) |>
    arrange(desc(within1kmfree))

bootstrapstat <- bootstrapgroup$mean[1] - bootstrapgroup$mean[2]
})
```

```r
# Visualize the bootstrap
bootstrap <- data.frame(bootstrap)
bootstrap |>
    ggplot(
        aes(x = bootstrap)
    )+ geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

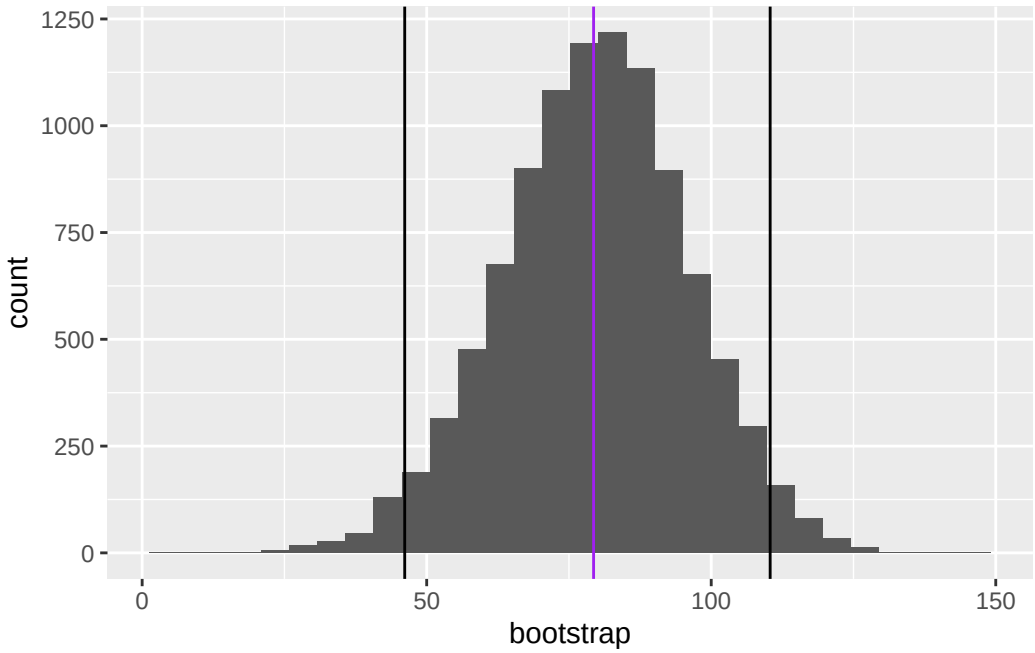



```
# Calculating 95% conf int
middle95 <- data.frame(quantile(bootstrap$bootstrap,probs=c(.025,.975)))
middle95
```

```
      quantile.bootstrap.bootstrap..probs...c.0.025..0.975..
2.5%                                                46.13715
97.5%                                              110.35345
```

```
# Visualizing with 95% conf int
bootstrap |>
    ggplot(
        aes(x = bootstrap)
    )+ geom_histogram()+
        geom_vline(xintercept = middle95[1,1])+
        geom_vline(xintercept = middle95[2,1]) +
        geom_vline(xintercept = sampstatfree, color = 'purple')
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



We would expect to find our estimate for difference in lead concentration between houses within 1km of a freeway and those further within our confidence interval(47.3814201347324-109.762161483208) 95% of the times we recreate this survey.

3. Simulate a negative result

Recall that a p-value is the probability of observing your sample *if* the null hypothesis is true. Therefore, if you draw a sample from two groups that part of the same population, you should get a p values less than 0.05 only 5% of the time. In other words, a negative result (no actual difference) will yield a significant p-value at the 0.05 threshold in 5% of samples. This is known as the **false positive rate**.

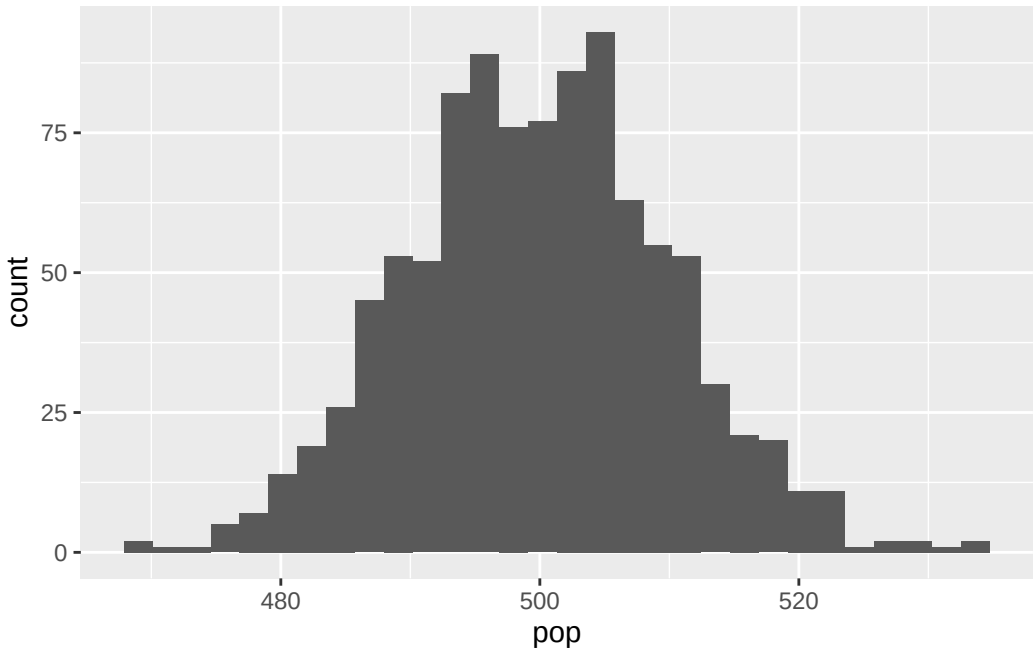
To simulate a negative result, perform the following.

1. Define a normally-distributed population. Pick any mean and standard deviation.
2. Draw a sample of 50 observations from this population. Assign the first 25 to group A and the second 25 to group B.
3. Conduct a permutation test to investigate whether the mean of group A is greater than the mean of group B.
4. Record your p-value.
5. Repeat the simulation 1,000 times.

```r
# 1 and 2
pop <- rnorm(1000, mean = 500, sd = 10)
popsamp <- sample(pop,50)
a <- head(popsamp,25)
b <- tail(popsamp,25)
a_b <- mean(a) - mean(b)
```

```r
# Visualizing the population our sample came from
pop |>
    data.frame() |>
    ggplot(
        aes(x = pop)
    )+ geom_histogram()
```

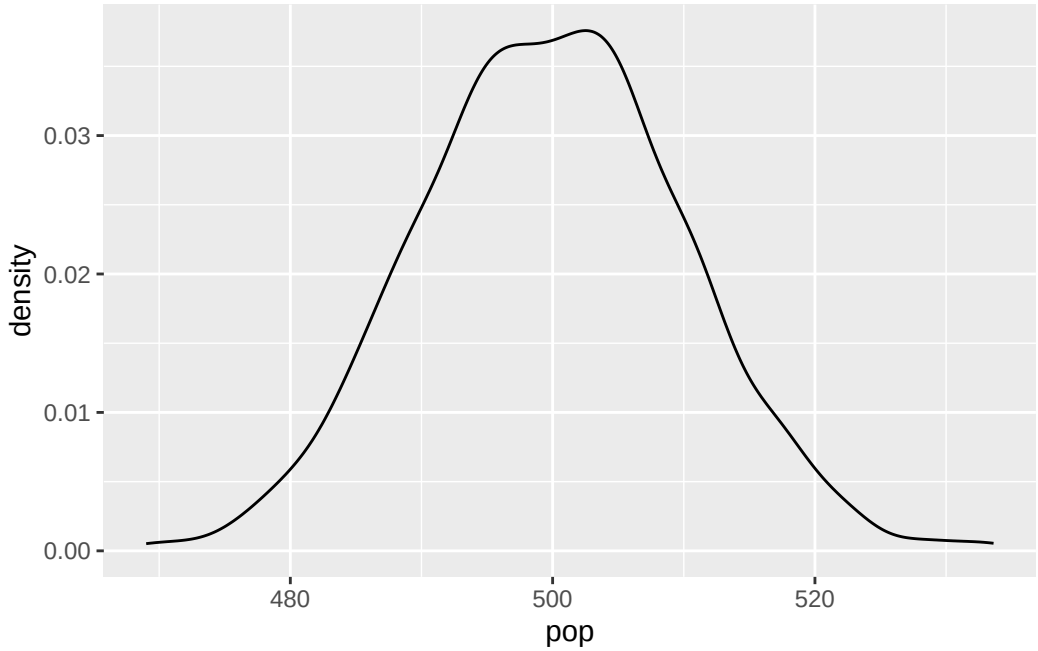
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
# Visualizing pdf of pop

pop |>
    data.frame() |>
    ggplot(
```

```
        aes(x = pop)
    )+
        geom_density()
```




```
# Make permutation
perm1 <- replicate(n = 10000, {
perma <- sample(pop,25)
permb <- sample(pop,25)
mean(perma) - mean(permb)
})
mean(perm1 > a_b)
```

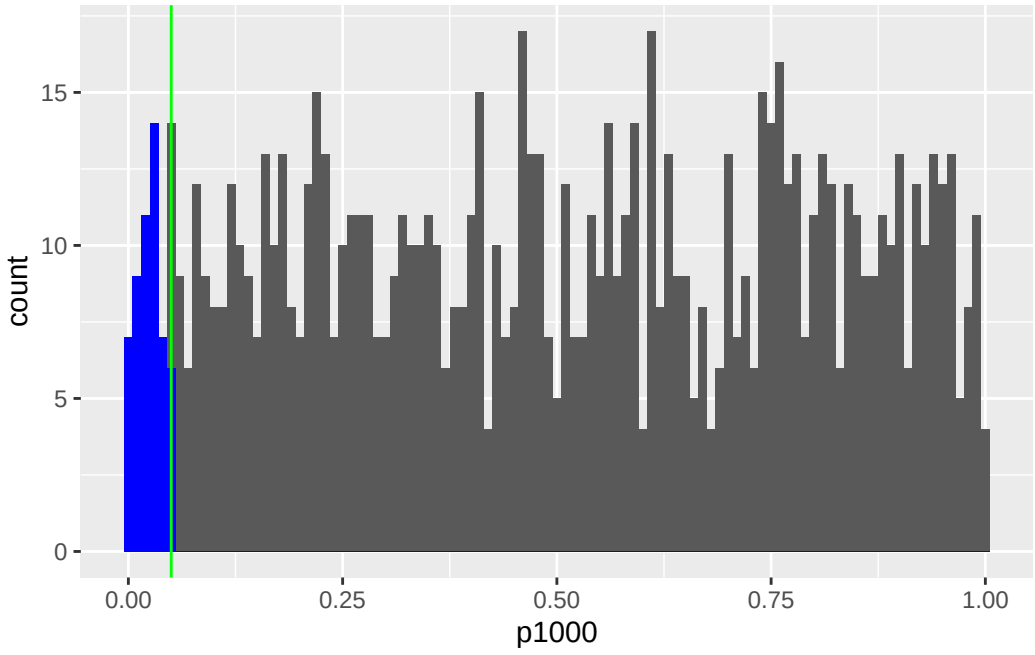
```
[1] 0.66
```

```
# Simulate 1000 times
p1000 <- replicate(n = 1000, {
    pop <- rnorm(1000, mean = 500, sd = 10)
popsamp <- sample(pop,50)
a <- head(popsamp,25)
b <- tail(popsamp,25)
a_b <- mean(a) - mean(b)
```

```
perm1 <- replicate(n = 10000, {
perma <- sample(pop,25)
permb <- sample(pop,25)
mean(perma) - mean(permb)
})
mean(perm1 > a_b)
})
```

```
# Visualizing our p-values

p1000 <- data.frame(p1000)
p1000 |>
    ggplot(aes(x = p1000))+
    geom_histogram(binwidth = 0.01)+
    geom_histogram(data = subset(p1000,p1000 <0.05),fill = "blue", binwidth = 0.01) +
    geom_vline(xintercept = 0.05, color='green')
```




```
# Verifying p
mean(p1000 < 0.05) # Is close to 0.05
```

```
[1] 0.054
```

```r
sum(p1000 < 0.05) # Is close to 50
```

```
[1] 54
```

When you're done, verify that your p-value is less than 0.05 in approximately 5% (50) of your simulations.